

Security+ 1.4 · Blockchain Technology

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 1.4** —
Blockchain Technology (2:22) · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Blockchain is a distributed public ledger where each block contains the hash of the previous block, creating a tamper-evident chain. Public blockchains (e.g., Bitcoin) are open and permissionless, while private blockchains restrict participation. Smart contracts are self-executing code deployed on the ledger.

High-Yield Core Concepts Tested on Exam

- Each block contains the hash of the PREVIOUS block — altering one breaks the chain
- Public blockchain (e.g., Bitcoin) is open; private blockchain requires permission
- Tamper-evident: any modification is detectable
- Smart contracts are code that runs on the blockchain
- Blockchain provides integrity and transparency, not confidentiality by default

Exam Tips & Classic Scenarios

- 'tamper-evident distributed ledger' = blockchain
- 'each block contains a hash of the previous' = proof of chaining
- If the scenario involves immutable logs or audit trails, think blockchain

Typical Exam Scenario: Scenario: 'A company wants to ensure that logs cannot be altered retroactively' — blockchain (tamper-evident).

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 1.4 — Blockchain Technology)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **Principle of Least Privilege:** Users and computational entities must be granted only the minimum permissions strictly required to perform designated tasks.
- **Defense-in-Depth (Layered Security):** Application of multiple security controls (technical, administrative, physical) to prevent single-point perimeter compromise.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. A characteristic of HASH (unlike encryption) is that it:

- A)** Can be reversed with the key
- B)** Is one-way (not reversible)
- C)** Always shortens the text by half
- D)** Requires a key pair

2. Which cryptographic function is NOT reversible?

- A)** AES encryption
- B)** RSA encryption
- C)** SHA-256 hash
- D)** Base64 encoding

3. Corporate scenario (Blockchain Technology): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 1.4), an analyst evaluates the following technical requirement: 'Each block contains the hash of the PREVIOUS one — change one, break the chain'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: Each block contains the hash of the PREVIOUS one — change one, break the chain...
- D)** Allow shared access among administrators using generic credentials

4. Corporate scenario (Blockchain Technology): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 1.4), an analyst evaluates the following technical requirement: 'Public (Bitcoin — open) vs Private (restricted permission)'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Public (Bitcoin — open) vs Private (restricted permission).
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

5. Corporate scenario (Blockchain Technology): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 1.4), an analyst evaluates the following technical requirement: '"tamper-evident distributed ledger' = blockchain.' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'tamper-evident distributed ledger' = blockchain...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

6. Corporate scenario (Blockchain Technology): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 1.4), an analyst evaluates the following technical requirement: 'each block contains a hash of the previous' = proof of chaining'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'each block contains a hash of the previous' = proof of chaining...
- D)** Allow shared access among administrators with generic credentials

7. Corporate scenario (Blockchain Technology): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 1.4), an analyst evaluates the following technical requirement: 'Scenario: company wants to ensure logs cannot be altered retroactively' — blockchain (tamper-evident). What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: Scenario: 'company wants to ensure logs cannot be altered retroactively' — blockchain (tamper-evident).
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Blockchain Technology): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 1.4), an analyst evaluates the following technical requirement: 'Each block contains the hash of the PREVIOUS one — change one, break the chain'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: Each block contains the hash of the PREVIOUS one — change one, break the chain...

9. Corporate scenario (Blockchain Technology): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 1.4), an analyst evaluates the following technical requirement: 'Public (Bitcoin — open) vs Private (restricted permission)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: Public (Bitcoin — open) vs Private (restricted permission)...

10. Corporate scenario (Blockchain Technology): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 1.4), an analyst evaluates the following technical requirement: 'tamper-evident distributed ledger' = blockchain'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: 'tamper-evident distributed ledger' = blockchain...

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) B — Is one-way (not reversible).

✓ **Why it is correct:** A fundamental characteristic of cryptographic hash functions is that they are one-way, i.e., not reversible. This means that from the hash, it is not possible to recover the original message. This property is known as pre-image resistance. Unlike encryption, which is reversible with the correct key, hashing is designed to be irreversible. CompTIA Security+ highlights that hash functions like SHA-256 are used for integrity and password storage (with salt), precisely because they cannot be reversed. The option 'Is one-way (not reversible)' correctly describes this essential characteristic.

✗ **Why the others are wrong:**

- *Can be reversed with the key:* This is a characteristic of symmetric or asymmetric encryption, not hash functions. In encryption, reversal is possible with the appropriate key (symmetric) or with the private key (asymmetric). Hash functions have no key and are not designed for reversal. Therefore, this statement is incorrect for hash. CompTIA Security+ emphasizes that hash is one-way and does not use keys, unlike cipher.
- *Always shortens the text by half:* Hash functions produce a fixed-size output regardless of input size. For example, SHA-256 always generates 256 bits (32 bytes). There is no 'shorten by half' rule; the digest size is constant and defined by the algorithm. Moreover, the hash function is not simple compression; it must be resistant to collisions and pre-image. This statement is technically false and does not describe a hash property.
- *Requires a key pair:* Hash functions do not use keys. They are public and deterministic algorithms that map input data to a digest. The use of a key pair is characteristic of asymmetric encryption (such as RSA or ECC), not hash. Although hashes can be used together with keys in HMAC, the hash function itself does not require keys. Therefore, this option is incorrect.

2) C — SHA-256 hash.

✓ **Why it's correct:** The correct answer is 'SHA-256 hash'. Cryptographic hash functions like SHA-256 are designed to be one-way (non-reversible). They produce a fixed-size digest (256 bits for SHA-256) from an input of any size. There is no secret key, and there is no practical way to reverse the hash to obtain the original message, except through brute-force attacks or precomputed tables, which are infeasible for complex inputs. CompTIA Security+ emphasizes that hashing ensures integrity, not confidentiality, and is irreversible by design.

✗ **Why the others are wrong:**

- *AES encryption:* AES (Advanced Encryption Standard) is a reversible symmetric encryption algorithm. With the correct key, the ciphertext can be decrypted back to plaintext. Therefore, it is incorrect to say AES is not reversible; its reversibility is a fundamental feature for confidentiality.
- *RSA encryption:* RSA is a reversible asymmetric encryption algorithm. The private key allows decryption of what was encrypted with the public key (and vice versa for signatures). Thus, RSA is reversible by nature, provided the appropriate key is known. This option is incorrect because it confuses encryption with hashing.
- *Base64 encoding:* Base64 is a data encoding that represents binary in ASCII text using 64 characters. It is not encryption and is fully reversible (decoding). It offers no security, only facilitates transport. This option is incorrect because Base64 is not a cryptographic function and is trivially reversible.

3) C — Implement the corresponding solution: Each block contains the hash of the PREVIOUS one — change one, break the chain....

✅ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 1.4 (Blockchain Technology): Each block contains the hash of the PREVIOUS one — change one, break the chain. This chaining ensures immutability and integrity; altering any block invalidates all subsequent blocks, providing tamper-evidence.

❌ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

4) A — Implement the corresponding solution: Public (Bitcoin — open) vs Private (restricted permission)..

✅ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 1.4 (Blockchain Technology): Public (Bitcoin — open) vs Private (restricted permission). Public blockchains are permissionless and open to anyone, while private blockchains are permissioned and restrict participation to authorized entities. This distinction is critical for security, privacy, and governance in blockchain deployments.

❌ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

5) B — Implement the corresponding solution: 'tamper-evident distributed ledger' = blockchain....

✅ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 1.4 (Blockchain Technology): 'tamper-evident distributed ledger' = blockchain. A blockchain is a decentralized, distributed ledger that uses cryptographic hashing and consensus mechanisms to make records tamper-evident, ensuring data integrity and immutability.

❌ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
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accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

6) C — Implement the corresponding solution: 'each block contains a hash of the previous' = proof of chaining....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 1.4 (Blockchain Technology): 'each block contains a hash of the previous' = proof of chaining.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

7) C — Implement the corresponding solution: Scenario: 'company wants to ensure logs cannot be altered retroactively' — blockchain (tamper-evident)..

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 1.4 (Blockchain Technology): Scenario: 'company wants to ensure logs cannot be altered retroactively' — blockchain (tamper-evident). Blockchain provides a tamper-evident ledger through cryptographic hashing and decentralized consensus, making it extremely difficult to alter historical records without detection. This property is valuable for ensuring log integrity and non-repudiation. Security+ includes blockchain as an emerging technology for enhancing data integrity and trust.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaches compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

8) D — Implement the corresponding solution: Each block contains the hash of the PREVIOUS one — change one, break the chain....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 1.4 (Blockchain Technology): Each block contains the hash of the PREVIOUS one — change one, break the chain.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).

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- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

9) D — Implement the corresponding solution: Public (Bitcoin — open) vs Private (restricted permission)....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 1.4 (Blockchain Technology): Public (Bitcoin — open) vs Private (restricted permission).

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
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10) D — Implement the corresponding solution: 'tamper-evident distributed ledger' = blockchain....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 1.4 (Blockchain Technology): 'tamper-evident distributed ledger' = blockchain.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
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